CCEE Preparation Cheat Sheet: Phase 1 (RDBMS & SQL)

CDAC DBDA Syllabus | Sessions 1–7

SESSION 1 & 2: CONCEPTS & ARCHITECTURE

File System vs. DBMS

- **File System:** High redundancy, low consistency, poor concurrency.
- **DBMS:** centralized control, supports ACID properties.
- **ACID Properties (Crucial for Transactions):**
  - **A**tomicity: All or nothing (Transaction fails? Rollback everything).
  - **C**onsistency: Database remains valid before and after transaction.
  - **I**solation: Concurrent transactions do not interfere with each other.
  - **D**urability: Once committed, data survives system crashes.

Codd's Rules (The "Big 3")

- **Rule 0 (Foundation):** The system must use relational facilities to manage the DB.
- **Rule 3 (Nulls):** Support for NULL values (distinct from 0 or empty string).
- **Rule 12 (Non-Subversion):** You cannot bypass integrity rules using low-level access.

Storage Architecture

Hierarchy: Database $\rightarrow$ Table Space (Logical) $\rightarrow$ Data File (Physical .dbf) $\rightarrow$ OS Block

- **Control File:** Binary metadata file (DB name, timestamps). Loss = DB won't start.

SESSION 3: SQL COMMANDS CLASSIFICATION

Type	Full Name	Key Commands	Behavior
DDL	Data Definition	`CREATE, ALTER, DROP, TRUNCATE`	Define Structure. **Auto-Commit**.
DML	Data Manipulation	`SELECT, INSERT, UPDATE, DELETE`	Manage Data. **Can Rollback**.
DCL	Data Control	`GRANT, REVOKE`	Permissions.
TCL	Transaction Control	`COMMIT, ROLLBACK`	Save/Undo DML changes.

Exam Trap: DELETE vs. TRUNCATE

- **DELETE (DML):** Row-by-row, Slower, Supports `WHERE`, Can `ROLLBACK`.
- **TRUNCATE (DDL):** Removes ALL rows, Faster (resets ID counters), No `WHERE`, **Cannot** `ROLLBACK`.

SESSION 4: QUERY LOGIC & ORDER OF EXECUTION

Execution Order (Memorize This!)

1. `FROM` (Load Tables)
2. `WHERE` (Filter Rows – *Cannot use Aliases here*)
3. `GROUP BY` (Bucket Data)
4. `HAVING` (Filter Groups – *Use for SUM, AVG, COUNT*)
5. `SELECT` (Choose Columns)
6. `ORDER BY` (Sort Results)

Joins Cheat Sheet

- **INNER JOIN:** Matches only. (No match? Row is dropped).
- **LEFT JOIN:** All from Left Table + Matches from Right (NULL if no match).
- **RIGHT JOIN:** All from Right Table + Matches from Left.
- **FULL OUTER JOIN:** Everything from both tables.

SESSION 5–7: DESIGN, NORMALIZATION & OBJECTS

Normalization (The "One-Liners")

- **1NF (Atomic):** One value per cell. (No comma-separated lists).
- **2NF (No Partial Dependency):** Non-key col must depend on the *whole* Primary Key. (Fix: Split table).
- **3NF (No Transitive Dependency):** Non-key $\rightarrow$ Non-key is forbidden. (e.g., City depends on Zip, not ID).

Database Objects

- **View:** A "saved query" or virtual table. Does not store data. Used for security (hiding columns).
- **Trigger:** Code that fires automatically on Event (INSERT/UPDATE/DELETE). Used for auditing/validation.
- **Stored Procedure:** Pre-compiled SQL code. Faster execution than raw SQL.